National University of Computer and Emerging

Artificial Intelligence Lab (AI 2002)

Date: March 17th, 2025

Course Instructor(s)
Ms. Ramsha Jat

Lab Mid Exam Paper A

Total Time: 90 minutes

Total Marks: 50

Total Questions: 03

Semester: Spring 2025

Campus: Karachi

Department: CS

Instructions:

- The Exam paper consists of 3 questions on 2 pages.
- Submit ipynb files name under your nu id with question number. E.g: k22-xxxx_Q1.ipynb.
- Submission will be on Software. Submission IP is **172.16.33.88:5000.**
- Enter your NU Id in Username and **Fast1234** in password.
- Read the question carefully before answering.

Student Name	Roll No	Section	Student Signature

LLO # 1: [18 Marks]

Q1. A firefighter agent must rescue a trapped person (P) in a burning building represented as a grid. The grid contains safe rooms (0), fire (F), walls (#), the agent's starting position (A), and the trapped person (P). The agent can only move through safe rooms (0) and must use DFS to find the shortest path to the person. Identify the type of agent and implement DFS would be implemented to solve this problem. For the example input map below, show the path taken to reach the treasure.

```
A 0 0 # #
# F 0 # P
0 0 0 F 0
0 # F # 0
0 0 0 0 0
```

LLO # 3: [18 Marks]

Q2. Implement an A search algorithm* to find the shortest path in an N × M grid, where each cell contains a numerical value representing its movement cost. Some cells are blocked (#), meaning they cannot be traversed. The algorithm should determine the least-cost path from a given starting position (S) to a target position (T) while considering the terrain costs. The movement is restricted to four directions: up, down, left, and right, with the cost of movement determined by the value of the destination cell. The heuristic function should use the Manhattan distance to estimate the remaining cost to the goal. The implementation should be able to handle larger grids efficiently and return both the optimal path and the total travel cost from the start position to the target position.

S	1	2	#	3
1	#	2	#	#
1	2	1	2	#
#	#	1	#	#
1	1	1	2	T

LLO # 4: [14 Marks]

Q3. Implement a Genetic Algorithm (GA) to find the maximum value of the function .
$f(x) = 2x + 1$
within the range [0, 31].
Each possible solution should be represented as a 5-bit binary string (since 25=322^5 = 3225=32, covering values from 0 to 31). The algorithm should begin with a random population of binary-encoded values and evaluate their fitness using $f(x) = x^2 + x$. It should then apply natural selection to retain the fittest individuals, followed by single-point crossover to generate new offspring. After crossover, each offspring should undergo single-bit mutation, where one randomly selected bit is flipped with a small probability. The process should continue for a fixed number of generations or until convergence, returning the best x value that maximizes f(x) along with the function's maximum value.